MIL-L-52906(ME)
30 August 1976

MILITARY SPECIFICATION

LOADER, SCOOP TYPE, DED, 4 x 4,

ARTICULATED FRAME STEER,

4-1/2 TO 5 CU. YD.,

COMMERCIaL CONSTRUCTION EQUIPMENT

(CCE)

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Amy, is available for use by all Departments and Agencies of the Department of Defense.

1.1 This specification covers a commercial model, scoop type, pneumatic-tired, four wheel drive, articulated frame steer, diesel-engine-driven, front-end loader.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-T-704 | Treatment and Painting of Matteriel. |
| MIL-L-2104 | Lubricating Oil, Internal Combustion Engine, Tactical Service. |
| MIL-L-2105 | Lubricating Oil, Gear, Multipurpose. |
| MIL-M-7866 | Molybdenum Disulfide, Technical, Lubrication Grade. |
| MIL-G-10924 | Grease, Automotive and Artillery. |
| MIL-G-23827 | Grease, Aircraft and Instrument, Gear and Actuator Screw. |

FSC 3805

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:39Z
Check the source to verify that this is the current version before use.

MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic.

STANDARDS

MILITARY

MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.

(Copies of specifications and standard required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications, The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

TECHNICAL INFORMATION PACKAGE

(Application for copies should be addressed to US Army Mobility Equipment Research and Development Command, ATTN: DRXFB-H, Fort Belvoir, VA 22060.)

DEPARTMENT OF LABOR (DOL)

29 CFR 1926 - Occupational Safety and Health Administration (OSHA) Standards,

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

3* REQIREMENTS

3.1 General. The loader shall be the latest model of the standard product of the contractor and shall have been marketed and in commercial field use in significant numbers for at least 1 year prior to the invitation



Source: https://assist.dla.mil -- Downloaded: 202610-22T21:39Z
Check the source to verify that this is the current version before use.

for bids or request for proposal. Product improvements incorporated in the commercial production units during this 1 year period are acceptable. When specific requirements are not stated herein, all items listed as standard equipment in the contractor's published specification brochures and catalogs, or normally furnished to commercial customers as standard equipment shall be furnished, The loader shall be equipped with any "--optional 'equipment necessary to comply with requirements specified herein. Optional equipment is defined as equipment not standard with the loader but which has been furnished to the commercial customer, such as special features or allied equipment. The loader shall be equipped with all components necessary to enable the loader to function reliably and efficiently in sustained operation. The loader shall conform to all Federal laws and regulations, applicable to the manufacturer, governing safety, noise levels, and pollution, which are in effect at the time of invitation for bids or request for proposal. For definition of terms used herein, SAE J732 shall apply.

3.2 Description. The loader shall be diesel-engine-driven, scoop type, pneumatic-tired, four-wheel-driven with rear axle oscillation, equipped with a hydraulically operated and front-mounted bucket (see 3.2.1.5).

3.2.1 Physical characteristics.

3.2.1.1 Weight. The loader shall weigh not less than 47,000 pounds. This weight shall include a fully serviced and fueled loader equipped with bucket, tires, and ROPS cab, as specified. The weight shall be determined without tire ballast and without removable counterweight.

3.2.1.2 Dump height. The loader shall have a dump height of not less than 115 inches measured to the cutting edge of the bucket at full lift and at a dump angle of 40 degrees.

3.2.1.3 Reach. The loader shall have a reach of not less than 42 inches at a dump height of 115 inches and at a dump angle of 40 degrees.

3.2.1.4 Dimensions. The loader shall have an overall length of not more than 360 inches. The overall width (exclusive of bucket) shall not exceed 126 inches.

3.2.1.5 Buckets. The loader shall be furnished with one of the following buckets as specified (see 6.2).



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

3.2.1.5.1 Rock bucket. The-bucket shall be a spade noise, “v" edge, or wedge” type rock bucket equipped with a spill plate. The bucket shall have an SAE nominal heaped rating of not less than 4-1/2 cubic yards in accordance with SAE J742. The bucket width shall not exceed 131 inches.

3.2.1.5.2 General purpose bucket. The bucket shall be the general purpose type. The bucket shall have an SAE nominal heaped rating of not less than 5 cubic yards in accordance with SAE J742. The bucket width shall not exceed 131 inches. Bolt-on type bucket teeth with replaceable tooth tips shall be furnished on the general purpose bucket. Number of teeth shall not be less than seven.

3.3. First article (preproduction model). The contractor shall furnish one loader with bucket(s) as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce loaders that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.2).

3.4 Production loaders. When loaders with rock buckets and loaders with general purpose buckets are procured under the same contract, the loaders shall be identical except for the bucket. All loaders furnished under this specification shall be physically and mechanically identical to the first article loader that is approved by the Government, except that product improvements incorporated in the contractor's standard commercial product lines are permitted at no cost subject to the following:

(a) Physical and performance characteristics have not been degraded.
(b) New components involving Government selected repair parts are directly interchangeable with the superseded component or part.
(c) The Government is notified of all changes and alterations a minimum of 45 days in advance of the production date of the change. When the changes and alterations have been incorporated in the production loaders, the Government shall be furnished the production effectivity by serial number and dates.
(d) Appropriate changes are issued to the Government validated operating, maintenance, and parts manuals before delivery.



Source: https://assist.dla.mil -- Downloaded: 202610-22T21:39Z
Check the source to verify that this is the current version before use.

D

3.5 Temperature. The loader shall start, without an external power source, and operate normally at full capacity for extended periods in temperatures ranging from -25° F to + 110° F, The loader shall be capable fo being stored for extended periods in temperatures ranging from -30°F to + 160°F without damage.

3.6 Use. The loader when equipped with a rock bucket (see 3.2.3.5.1), shall be capable of efficient and reliable operation under conditions normally encountered in the loading of blast rock in quarry operations. The loaders when equipped with a general purpose bucket (see 3,2.1.5.2), shall be capable of efficient and reliable operation under conditions normally encountered in excavating, digging, loading, strippings leveling, stockpiling, and transferring of earth and aggregate.

3.7 Safety.

3.7.1 Roll-Over Protective Structure (ROPS). The loader shall be equipped with a ROPS conforming to SAE J394. A seatbelt conforming to SAE J386 shall be furnished.

3.7.2 Falling Object Protective Structure (FOPS). The loader shall be equipped with a FOPS conforming to SAE J231,

3.7.3 Sound level. 'The sound level of the loader shall not exceed the OSHA noise level exposure limits in effect on the date of invitation for bids or request for proposal. If the sound level of the loader exceeds 85dB(A) when measured in-accordance with SAE J919, a warning plate or decal containing the following shall be permanently affixed to the loader in a conspicuous, protected location in the operator's compartment: "Warning, Ear Protection Required When Operating This Loader."

3.7.4 Horn. A hand- or foot-operated horn shall be furnished.

3.7.5 Backup alarm. A backup alarm conforming to SAE J994 shall be furnished.

3.7.6 Neutral start. A neutral start mechanism shall be furnished and shall prevent the starter from cranking the engine when the transmission is in any gear.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

3,8 Performance.

3.8.1 Operating load. The loader shall have an SAE rated operating load of not less than 3,000 pounds per cubic yard of the nominal heaped bucket rating (reference SAE J732, J742, J818),

3.8.2 Tipping load. The loader shall have a tipping load, SAE rating, of not less than twice the rated operating load (reference SAE J732).

3.8.3 Lift capacity. The loader shall have an SAE lift capacity at the carry position of not less than 42,000 pounds (reference SAE 3732).

3.8.4 Breakout force. The loader shall have an SAE breakout force of not less than 26,000 pounds (reference SAE J732).

3.8.5 Raising time. The loader shall have an SAE raising time of not more than 8.5 seconds (reference SAE J732).

3.8.6 Travel speed. The empty loader shall be capable of a forward travel speed of not less than 19 mph on level pavement.

3.8.7 Drawbar pull. The empty loader shall develop a drawbar pull of not less than 25,000 pounds at 1.5 mph, at the specified operating weight, on a dry, level, concrete surface, with not more than 15 percent slip factor.

3.8.8 Clearance circle. The loader shall have a curb clearance circle of not more than 50 feet (reference SAE J695),

3.9 Diesel engine. The diesel engine shall develop not less than 260 net flywheel horsepower at the rated governed speed of not more than 2,200 rpm and at the SAE standard ambient conditions (reference SAE J816). The engine lubricating system shall maintain operating pressure on 30 percent side slopes and 60 percent longitudinal slopes. An air cleaner restriction gage shall be furnished.

3.9.1 Engine side panels or rear fenders. Removable engine side panels or rear fenders shall be provided to prevent water, mud and debris from being thrown into the engine compartment.

3.10 Torque converter and transmission. The loader shall be furnished with a torque converter, a full power shift transmission, and a torque



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

converter oil cooler. The transmission shall provide not less than four forward and three reverse speeds; or, when a dual-range twin turbine torque converter is furnished, not less than two forward and two reverse speeds.

3.11 Braking systems. A foot controlled service braking system shall be provided on all four wheels. A parking system and an emergency stopping system shall be provided. The braking systems shall comply with SAE J237.

3.12 Steering. The loader shall have articulated frame steering with a minimum of 35 degrees of articulation in each direction,

3.13 Tires. Tires shall be of the deep rock type tread (L-4), nylon, tubeless, with not less than a 20 ply rating. Tires shall be of the wide 'base type with a size of not less than 26.5 x 25. All tires shall be of the same size, type, and ply rating. Sufficient clearance shall be provided to mount 26.5 x 25, 20 ply rating, extra deep rock tread (L-5) type tires without removal of any components.

3.14 Electrical system. A 24-volt electrical system shall be provided.

3.14.1 Alternator. An alternator shall be provided and shall have not less than a 45-amp rating.

3.14.2 Instruments, The following instruments shall be provided:

(a) Engine coolant temperature gage.
(b) Ammeter or voltmeter.
(c) Engine oil pressure gage.
(d] Torque converter oil temperature gage.
(e) Air pressure gage.
(f) Fuel level indicator (dipstick or gage).
(g) Engine hour meter.

3.14.3 Lighting. Not less than four front work/travel lights, two rear work lights, two taillights, and two stoplights shall be furnished.

3.15 Boom and bucket controls. An adjustable automatic boom kickout and an adjustable automatic bucket positioner shall be furnished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

3.16 Cab. A fully enclosed cab with door(s), windows, and front and rear windshield shall be furnished. A heater and a defroster shall be furnished. A front windshield wiper(s) shall be furnished.

3.17 Counterweight and tire ballast. Counterweight may be furnished on the loader to improve stability and to meet the requirements (except 3.2,1.1, Weight) of this specification, Tire ballast is not acceptable and shall not be furnished.

3.18 Fuel capacity. The loader shall be capable of operating for a minimum of 5 hours under full engine load without refueling, as calculated by the contractor.

3.19 Drawbar. The loader shall be equipped with a fixed type drawbar including a drawbar pin and a positive pin retaining means.

3.20 Toolbox. A toolbox shall be furnished.

3.21 Transportability. The loader shall be equipped with lifting eyes and shall be capable of being lifted without damage using slings, and spreader bars if needed. The loader shall be equipped with tiedown eyes and shall be capable of being transported by surface, rail, and C-5A aircraft. Cargo envelope of C-5A aircraft is shown in Figure 1. Complete diagrams and instructions for lifting the loader and tying down the loader on freight cars and truck trailers shall be furnished in the operator's manual or a supplement to the operator's manual. The operators manual or manual supplement shall include instructions for component removal when required for transport, Instruction plates or decals containing diagrams and instructions for use of the lifting and tiedown provisions shall be permanently affixed to the loader. Stencil or decal markings shall be applied to the loader at each lifting and tiedown point. A steering frame lock in accordance with SAE J276 shall be provided.

3.22 Identification plates and marking. Each loader shall be furnished with an identification plate showing the manufacturer, model numbers, NSN, serial number, date of manufacture, and date of delivery. The identification plate shall be permanently affixed to the loader in a readily accessible and conspicuous location, Instruction or warning plates or decals shall be furnished and mounted in accordance with normal commercial practice.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

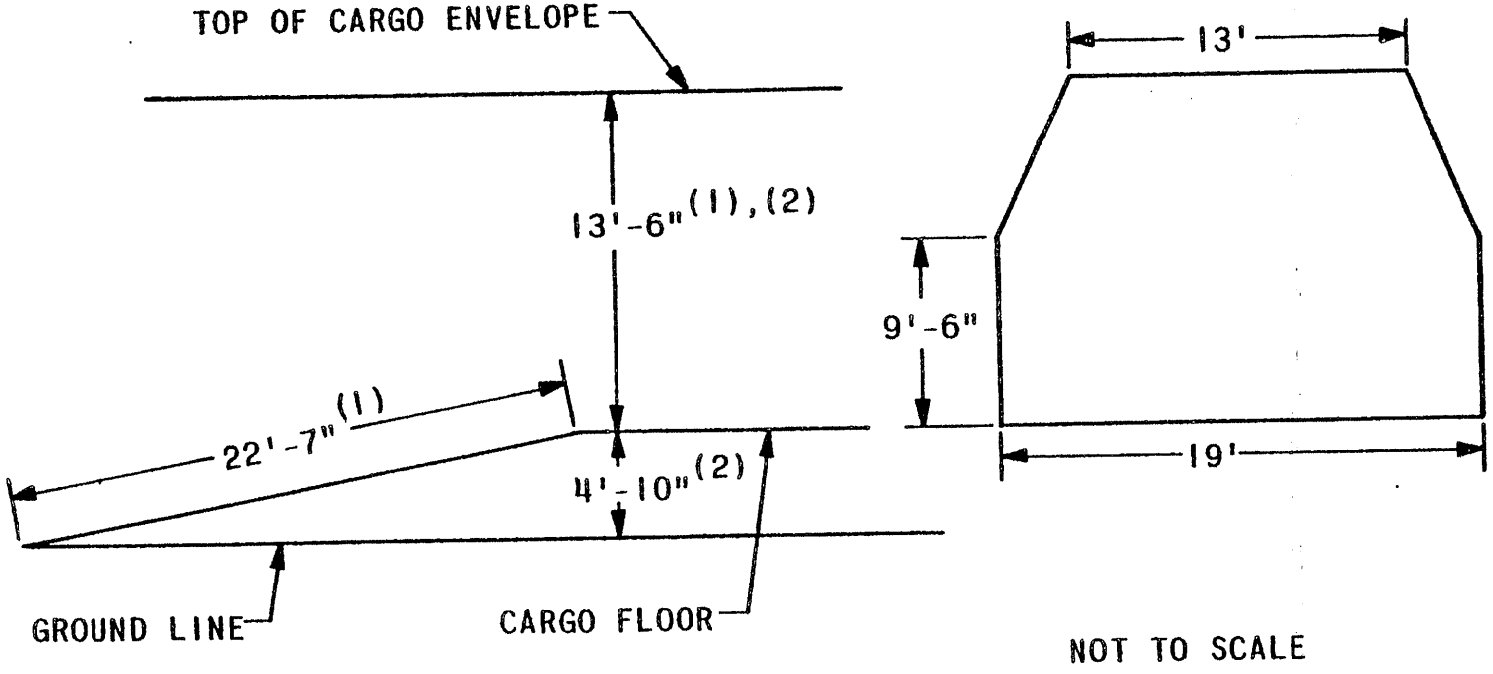



FIGURE 1. Forward cargo opening C-5A forward kneel position.

X-3162 A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

3.23 Treatment and painting. The loader shall be cleaned, treated, and painted in accordance with the contractors standard commercial practice. The final color shall be lusterless forest green in accordance with MIL-T-704.

3.24 Marking. The Army registration number and the operating weight of the loader shall be stenciled on both sides of the loader in such a manner as to be readily identifiable. Tire pressure shall be stenciled on the loader adjacent to each tire. All marking shall be in accordance with MIL-STD-642. Color of the marking shall be black.

3.25 Lubricants. The loaders shall be serviced with the following lubricants:

Engine: MIL-L-2304 MIL-L-46167
Torque converter: MIL-L-2104 or MIL-L-46167
Transmission: MIL-L-2104 or MIL-L-46167
Hydraulic system: MIL-L-2104 or MIL-L-46167.
Differentials and planetaries: MIL-L-2105
Grease lubrication points: MIL-G-10924, MIL-G-23827 or MIL-M-7866

Lubricants not specified shall be standard commercial lubricants.

3.26 Techical publications. Such technical publications as are specified (see 6.2) shall be furnished.

3.27 Techincal, information package. The contractor shall complete and submit a technical information package within the time frame specified (see 6.2). The information shall depict the complete loader and its components exactly as it will be furnished. Notation shall be made of all components which have not been in field use for a minimum of 1 year, and shall include identification of the component or part previously used.

3.28 Government-furnished equipment. The following item shall be furnished by the Government. (see 6.3):

(a) Case, maintenance and operator's manuals, NSN 7520-00-559-9618. Logbook, binder, and forms. The contractor shall furnish the fasteners and shall install the case on the loader.
(b) Vehicle classification sign kit, NSN 9905-00-565-6267. The contractor shall furnish the fasteners and install the kit on the loader.



Source: https://assist.dla.mil -- Downloaded: 202610-22T21:39Z
Check the source to verify that this is the current version before use.

(c) Materiel fielding plan.
(d) Prescribed Load List (PLL) and Authorized Stockage List (ASL).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Goverment. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3),
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection. Prior to the examination and test of the loaders the following shall be performed:

(a) Service the loader with the lubricants specified.
(b) Limber up the loader as prescribed by the contractor,

4.3.1 Preproduction examination. Prior to testing, the completed technical information package shall be reviewed. Any discrepancies between the completed technical information package, the specification, and the preproduction model shall be cause for rejection. Prior to testing, the loader shall be examined for defects marked "X" in Column 1 of Table I. Presence of one or more defects shall be cause for rejection. Prior to testing, the calculations, certifications, and certified test reports required by 4.5.2.13, 4,5.2.14, and 4.5.2.15 shall be reviewed. Incomplete, inaccurate, or otherwise unacceptable calculations, certifications, or certified test reports shall be cause for rejection.

4.3.2 Preproduction test. Upon successful completion of the examination specified in 4.3.1, the preproduction model loader shall be subjected to the tests marked "X" in Column 1 of Table II. When a single loader furnished with both a rock bucket and a general purpose bucket is scheduled



Source: https://assist.dla.mil -- Downloaded: 202610-22T21:39Z
Check the source to verify that this is the current version before use.

for preproduction model test, the tests for weight, dump height, reach, tipping load, lift capacity, breakout forces and raising time shall be conducted on both the loader equipped with the rock bucket and on the same loader equipped with the general purpose bucket. The tests for production run-in, travel speed, drawbar pulls clearance circle, braking systems, and lifting provisions may be conducted on either the loader equipped with the rock bucket or the general purpose bucket. The production run-in test shall be the first test conducted. Failure of any test shall be cause for rejection.

Table I. Examination Schedule

| Preproduction Model | Individual | Defect | | Requirement Paragraph |
|---|---|---|---|---|
| 1 | 2 | 3 | | 4 |
| X | X | 101. | Overall length not as specified. | 3.2.1.4 |
| X | X | 102. | Overall width not as specified. | 3.2.1.4 |
| X | X | 103. | Bucket not as specified. | 3.2.1.5.1, 3.2.1.5.2 |
| X | X | 104. | ROPS not as specified. | 3.7.1 |
| X | X | 105. | FOPS not as specified. | 3.7.2 |
| X | X | 106. | Sound level warning plate not as specified. | 3.7.3 |
| X | X | 107. | Horn not as specified. | 3.7.4 |
| X | X | 108. | Backup alarm not as specified. | 3.7.5 |
| X | X | 109 | Neutral start not as specified. | 3.7.6 |
| X | X | 110. | Diesel engine not as specified. | 3.9 |
| X | X | 111. | Engine side panels or rear fenders not as specified. | 3.9.1 |
| X | X | 112. | Torque converter not as specified. | 3.10 |



Source: https://assist.dla.mil -- Downloaded: 202610-22T21:39Z
Check the source to verify that this is the current version before use.

Table I Examination Schedule (Continued)

| 1 | 2 | 3 | | 4 |
|---|---|---|---|---|
| X | X | 113. | Torque converter oil cooler not as specified. | 3.10 |
| X | X | 114. | Transmission not as specified. | 3.10 |
| X | X | 115. | Braking system not as specified. | 3.11 |
| X | X | 116. | Steering system not as specified. | 3.12 |
| X | X | 117. | Tires not as specified. | 3.13 |
| X | X | 118. | Insufficient clearance to mount L-5 tires. | 3.13 |
| X | X | 119. | Electrical system not as specified. | 3.14 |
| X | X | 120. | Alternator not as specified. | 3.14.1 |
| X | X | 121. | Instruments not as specified. | 3.14.2 |
| X | X | 122. | Lighting not as specified. | 3.14.3 |
| X | X | 123. | Boom and bucket controls not as specified. | 3.15 |
| X | X | 124. | Cab not as specified. | 3.16 |
| X | X | 125. | Drawbar not as specified. | 3.19 |
| X | X | 126. | Toolbox not as specified. | 3.20 |
| X | X | 127. | Transportability not as specified. | 3.21 |
| X | X | 128. | Steering frame lock not as specified. | 3.21 |
| X | X | 129. | Identification plates and marking not as specified. | 3.22 |
| X | X | 130. | Paint not as specified. | 3.23 |
| X | X | 131. | Stenciling not as specified. | 3.24 |
| X | X | 132. | Lubricants not as specified. | 3.25. |
| X | X | 133. | Parts or components missing. | Section 3 |
| X | X | 134. | Assembly incomplete or incorrect. | Section 3 |
| X | X | 135. | Cracked or broken parts or assemblies. | Section 3 |
| X | X | 136. | Permanent deformation of any part or assembly. | Section 3 |
| X | X | 137. | Leaks in coolant, fuel, oil, and air lines. | Section 3 |
| X | X | 138. | Mechanical failure or deterioration of insulation. | Section 3 |
| - | X | 139. | Material not of same quality as used on preproduction model. | Section 3 |
| X | X | 140. | Loader not as specified. | Section 3 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

Table II. Test schedule

| Preproduction Model | Individual | Test | Test Paragraph | Requirement Paragraph |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| X | - | Weight. | 4.5.2.1 | 3.2.1.1 |
| X | - | Dump height. | 4.5.2.2 | 3.2.1.2 |
| X | - | Reach. | 4.5.2.3 | 3.2.1.3 |
| X | - | Tipping load. | 4.5.2.4 | 3.8.1, 3.8.2 |
| X | - | Lift capacity. | 4.5.2.5 | 3.8.3 |
| X | - | Breakout force. | 4.5.2.6 | 3.8.4 |
| X | - | Raising time. | 4.5.2.7 | 3.8.5 |
| X | - | Travel speed. | 4.5.2.8 | 3.8.6 |
| X | - | Drawbar pull. | 4.5.2.9 | 3.8.7 |
| X | - | Clearance circle. | 4.5.2.10 | 3.8.8 |
| X | - | Braking systems. | 4.5.2.11 | 3.11 |
| X | - | Lifting provisions. | 4.5.2.12 | 3.21 |
| X | X | Production run-in. | 4.5.2.16 | Section 3 |

4.4 Quality conformance inspection.

4.4.1 Individual examination. After successful completion of the test specified in 4.4.2, each loader shall be examined for the defects marked "X" in Column 2 of Table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Individual test. Each loader shall be tested as specified in Column 2 of Table II. Failure of any test shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The examination shall be in accordance with Table I.



Source: https://assist.dla.mil -- Downloaded: 202610-22T21:39Z
Check the source to verify that this is the current version before use.

4.5.2 Tests. Tests shall be conducted in accordance with Table II.

4.5.2.1 Weight. The weight of the loader shall be measured with a scale calibrated within the last 6 months to prove conformance to 3.2.1.1. Nonconformance to 3.2.1.1 shall be cause for rejection.

4.5.2.2 Dump height. The dump height of the loader shall be measured to prove conformance to 3.2.1.2. Nonconformance to 3.2.1.2 shall be cause for rejection,

4.5.2.3 Reach. The reach of the loader shall be measured to prove conformance to 3.2.1.3. Nonconformance to 3.2.1.3 shall be cause for rejection.

4,.5.2.4 Tipping load. The tipping load of the loader shall be measured to prove conformance to 3.8.1 and 3.8.2. Nonconformance to 3.8.1 and 3.8.2 shall be cause for rejection.

4.5.2.5 Lift capacity. The lift capacity of the loader shall be measured to prove conformance to 3.8.3. Nonconformance to 3.8.3 shall be cause for rejection.

4.5.2.6 Breakout force. The breakout force of the loader shall be measured to prove conformance to 3.8.4. Nonconformance to 3.8.4 shall be cause for rejection.

4.5.2.7 Raising time. The raising time of the loader shall be measured to prove conformance to 3.8.5. Nonconformance to 3.8.5 shall be cause for rejection.

4.5.2.8 Travel speed. The travel speed of the loader shall be measured to prove conformance to 3.8.6. Nonconformance to 3.8,6 shall be cause for rejection.

4.5.2.9 Drawbar pull. The drawbar pull of the loader shall be measured to prove conformance to 3.8.7. Nonconformance to 3.8.7 shall be cause for rejection.

4.5.2.10 Clearance circle. The curb clearance circle shall be measured to prove conformance to 3.8.8. Nonconformance to 3.8.8 shall be cause for rejection.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

4.5.2.11 Braking systems. The braking systems of the loader shall be tested to prove conformance to 3.11, Nonconformance to 3.11 shall be cause for rejection,

4,5.2.12 Lifting previsions. Lift the loader using slings, and spreader bars if needed, to prove conformance to 3.21. Nonconformance to 3.21 shall be cause for rejection.

4.5.2.13 Calculations. The contractor shall furnish calculations proving compliance with the bucket capacity requirements of 3.2,1.5.1 or 3.2.1.5.2, as applicable. Dimensions used in the calculations shall be verified by measurement. The contractor shall furnish calculations proving compliance with the fuel capacity requirements of 3.18, Nonconformance to 3.2.1.5.1, 3.2.1.5.2, or 3.18 shall be cause for rejection.

4.5.2.14 Certification. The contractor shall certify that the loader complies with temperature requirements of 3.5. Nonconformance to 3.5 shall be cause for rejection. The contractor shall certify that the loader complies with the sound level requirements of 3.7.3. Nonconformance to 3.7.3 shall be cause for rejection.

4,5.2.15 Certified test reports. The contractor shall furnish certified test reports to prove compliance with the ROPS requirements of 3.7.1, and the FOPS requirements of 3.7.2. If certified test reports are not available the ROPS and FOPS shall be tested to prove conformance to 3.7.1 and 3.7.2, respectively. Nonconformance to 3.7.1 or 3.7.2 shall be cause for rejection.

4.5.2.16 Production run-in. Each loader shall be operated as follows: Operate the loader for not less than 1/2 hour with the bucket empty in all gears, forward and reverse, at maximum speeds commensurate with conditions Include not less than three starts and stops in each gear. Steer the loader to full lock, either moving or static, not less than six times in each direction. Stop the engine and restart it. Operate the hydraulic controls for not less than 1/2 hour. During this 1/2 hour operation, raise and lower the empty bucket through the full lifting cycle at least five times. Operate the bucket from full rollback to full dump at least five times. Hold each hydraulic control in each actuation position until the hydraulic system reaches relief valve pressure at least five times. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any control, or evidence of leakage of fuel, lubricant, or hydraulic oil



Source: https://assist.dla.mil -- Downloaded: 202610-22T21:39Z
Check the source to verify that this is the current version before use.

shall be cause for rejection of the loader by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after the deficiency is corrected.

5. PACKAGING

5.1 Preservation, packing, and marking. Each complete loader, and all items issued with the loader, shall be suitably preserved and packed for retail distribution directly to a using customer in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with carrier rules and regulations applicable to the mode of transportation In addition, the following specific requirements shall " be accomplished. Items with critical surfaces required for fit or ' . function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors, or dessicated packs. Items requiring protection from physical damage shall be protected by wrapping, pack compartmentization, or carbonizing of the individual item. The cooling system shall be filled with a fresh, clean solution of 50 percent water and 50 percent ethylene glycol type antifreeze, and a tag shall be attached indicating the temperature to which the cooling system can be subjected before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating the preservation and packing materials to be removed in detail, and any other servicing required prior to placing the loader in operation. Marking shall reflect, as a minimum, the item name, national stock number, quantity, contract or purchase order number, and appropriate address markings on each unpacked item or shipping container The marking shall be applied with a waterproof material, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The loader is intended for loading blast rock in quarry operations, excavating, digging, loading, stripping> leveling, stockpiling, and transferring of earth and aggregate.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type bucket required (see 3.2.1.5).
(c) Time frame required for submission of the preproduction model (see 3.3).
{d) Technical publications required (see 3.26).
(e) Time frame for completing and submitting a technical information package (see 3.27).



Source: https://assist.dla.mil -- Downloaded: 202610-22T21:39Z
Check the source to verify that this is the current version before use.

6.3 Government-furnished equipment. The contracting officer should arrange to furnish the equipment specified in 3.28.

6.4 Preproduction models When only loaders equipped with rock buckets are being procured, the preproduction test should be conducted with a loader equipped with a rock bucket. When only loaders equipped with general purpose buckets are being procured, the preproduction test should be conducted with a loader equipped with a general purpose buckets When both loaders equipped with rock buckets and loaders equipped with general purpose buckets are being procured, the preproduction test should be conducted with a single loader furnished with both a rock bucket and a general purpose bucket.

Custodian:
Army - ME

User activity:
Army - AT

Preparing activity:
Army - ME

Project 3805-A089



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-L-52906(ME) Loader, Scoop Type, DED, 4 x 4, Articulated Frame Steer, 4-½ to 5 Cu.Yd., Commercial Construction Equip

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:39Z
Check the source to verify that this is the current version before use.